

Problem Set

- Q1.** A die is thrown. Find $P(\text{Even} \mid \text{Number} > 3)$.
- Q2.** A card is drawn. Find $P(\text{King} \mid \text{Face card})$.
- Q3.** Two coins are tossed. Given at least one head, find the probability of two heads.
- Q4.** A bag has 3 red and 2 blue balls. Two balls are drawn without replacement. Find the probability of drawing both balls red.
- Q5.** If $P(A) = 0.6$, $P(B) = 0.5$ and $P(A \cap B) = 0.3$, find $P(A \mid B)$.
- Q6.** A bag has 4 green and 6 yellow balls. Two balls are drawn with replacement. Find the probability of drawing both green balls.
- Q7.** Machine A produces 70% items (4% defective). Machine B produces 30% items (5% defective). Find the probability of an item being defective.
- Q8.** Using Q8 data, find the probability that a defective item came from Machine A.
- Q9.** Probability of rain = 0.3. If it rains, the probability of a traffic jam is $P(J) = 0.8$. If there's no rain, the probability of a traffic jam is 0.4. Find the probability of traffic jam.
- Q10.** In Q9, find the probability that it rained, given a traffic jam.
- Q11.** A test is 95% accurate. Prevalence of disease = 2%. Find the probability person actually has the disease given a positive result.
- Q12.** Two cards are drawn without replacement. Find the probability second card is an ace given first is an ace.
- Q13.** In a class, 60% pass Maths, 70% pass English, 50% pass both. Find the probability that student passes at least one subject.
- Q14.** A box has 5 white and 7 black balls. Two balls are drawn without replacement. Find the probability of exactly one white ball.
- Q15.** If $P(A) = 0.4$, $P(B) = 0.5$ and A,B independent, find $P(A \cup B)$.
- Q16.** Three bulbs are selected from 5 good and 3 defective. Find the probability that all are good.
- Q17.** A factory has three plants A (50%), B (30%), C (20%). Defective rates: 2%, 3%, 4% respectively. Find the probability item is defective.
- Q18.** Using Q17 data, find the probability that a defective item came from Plant C.

Solutions

S1. Numbers $> 3 = \{4,5,6\}$, Even within these $= \{4,6\}$, so $P = 2/3$.

Hint (formula): $P(A | B) = |A \cap B| / |B|$.

S2. Face cards = 12, Kings = 4, so $P = 4/12 = 1/3$.

Hint (formula): $P(A | B) = P(A \cap B) / P(B)$.

S3. Reduced sample space $\{HH, HT, TH\}$, so probability $= 1/3$.

Hint (formula): $P(A | B) = P(A \cap B) / P(B)$.

S4. $(3/5) \times (1/4) = 3/20$.

Hint (formula): $P(A \cap B) = P(A)P(B | A)$.

S5. $P(A | B) = 0.3/0.5 = 0.6$.

Hint (formula): $P(A | B) = P(A \cap B) / P(B)$.

S6. With replacement (independent): $(4/10) \times (4/10) = 4/25$.

Hint (formula): If independent, $P(A \cap B) = P(A)P(B)$.

S7. $P(D) = (0.7)(0.04) + (0.3)(0.05) = 0.028 + 0.015 = 0.043$.

Hint (formula): Total probability: $P(D) = \sum (P(D | A_i)P(A_i))$.

S8. $P(A | D) = (0.7 \cdot 0.04) / 0.043 \approx 0.651$.

Hint (formula): Bayes theorem

S9. $P(J) = (0.3)(0.8) + (0.7)(0.4) = 0.24 + 0.28 = 0.52$.

Hint (formula): $P(J) = P(J | R)P(R) + P(J | R^c)P(R^c)$.

S10. $P(R | J) = 0.462$

S11. $P(T) = (0.02)(0.95) + (0.98)(0.05) = 0.019 + 0.049 = 0.068$,

$P(D | T) = 0.019 / 0.068 \approx 0.279$.

Hint (formula): $P(D | T) = (P(T | D)P(D)) / P(T)$.

S12. After first ace, remaining aces = 3 out of 51, so $P = 3/51 = 1/17$.

Hint (formula): Conditional probability with reduced sample space.

S13. $P(M \cup E) = 0.6 + 0.7 - 0.5 = 0.8$.

Hint (formula): $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

S14. $(5/12) \times (7/11) + (7/12) \times (5/11) = 35/66$.

Hint (formula): Exactly one success: $P(WB) + P(BW)$.

S15. $P(A \cup B) = 0.4 + 0.5 - (0.4)(0.5) = 0.9 - 0.2 = 0.7$.

Hint (formula): Independence: $P(A \cap B) = P(A)P(B)$ and $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

S16. $(5/8) * (4/7) * (3/6) = 5/28$.

Hint (formula): Without replacement: multiply conditional probabilities.

S17. $P(D) = (0.5)(0.02) + (0.3)(0.03) + (0.2)(0.04) = 0.01 + 0.009 + 0.008 = 0.027$.

Hint (formula): $P(D) = \sum P(D | i)P(i)$.

S18. $P(C | D) = ((0.2)(0.04)) / 0.027 \approx 0.296$.

Hint (formula): Bayes: $P(C | D) = (P(D | C)P(C)) / P(D)$